

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital VLSI Testing (01CT0629)	Aim: To verify the truth tables of different logic gates.	
Experiment No: 4	Date:04/03/2026	Enrolment No: 92301733056

AIM: To verify the truth tables of different logic gates.

Software/Platform required: EDA Playground, Quartus.

THEORY:

Logic gates are the basic building blocks of digital electronics. They perform logical operations on one or more binary inputs (0 or 1) to produce a single binary output. The relationship between input combinations and the corresponding output is represented using a truth table.

Each logic gate performs a specific function:

- AND gate: Output is 1 only when all inputs are 1.
- OR gate: Output is 1 if at least one input is 1.
- NOT gate: Produces the complement (inversion) of the input.
- NAND gate: Complement of AND gate output.
- NOR gate: Complement of OR gate output.
- XOR (Exclusive-OR) gate: Output is 1 when inputs are different.
- XNOR gate: Output is 1 when inputs are the same.

By applying all possible input combinations and observing the outputs, we can verify whether the gate follows its theoretical truth table. This verification ensures correct functioning of digital circuits and helps in understanding the logical behavior of each gate.

Thus, truth table verification is essential for analyzing and designing reliable combinational logic circuits.

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VERILOG CODE:

A. Logic Gates Module

```

module gates (
    input wire A,
    input wire B,
    output wire AND_OUT,
    output wire OR_OUT,
    output wire NOT_OUT,
    output wire NAND_OUT,
    output wire NOR_OUT,
    output wire XOR_OUT,
    output wire XNOR_OUT
);
    assign AND_OUT = A & B;
    assign OR_OUT = A | B;
    assign NOT_OUT = ~A;
    assign NAND_OUT = ~(A & B);
    assign NOR_OUT = ~(A | B);
    assign XOR_OUT = A ^ B;
    assign XNOR_OUT = ~(A ^ B);
endmodule

```

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B. Testbench

```

`timescale 1ns/1ps

module testbench;

// Testbench signals
reg A;
reg B;

wire AND_OUT;
wire OR_OUT;
wire NOT_OUT;
wire NAND_OUT;
wire NOR_OUT;
wire XOR_OUT;
wire XNOR_OUT;

// Instantiate the DUT (Device Under Test)
gates uut (
.A(A),
.B(B),
.AND_OUT(AND_OUT),
.OR_OUT(OR_OUT),
.NOT_OUT(NOT_OUT),
.NAND_OUT(NAND_OUT),
.NOR_OUT(NOR_OUT),
.XOR_OUT(XOR_OUT),
.XNOR_OUT(XNOR_OUT)
);

initial begin

// Create VCD file for waveform
$dumpfile("dump.vcd");
$dumpvars(0, testbench);

$display("A B | AND OR NOT NAND NOR XOR XNOR");
$display("-----");

// Apply test vectors
A = 0; B = 0; #10;
$display("%b %b | %b %b %b %b %b %b %b",
A,B,AND_OUT,OR_OUT,NOT_OUT,NAND_OUT,NOR_OUT,XOR_OUT,XNOR_OUT);

A = 0; B = 1; #10;
$display("%b %b | %b %b %b %b %b %b %b",
A,B,AND_OUT,OR_OUT,NOT_OUT,NAND_OUT,NOR_OUT,XOR_OUT,XNOR_OUT);

```

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```

A = 1; B = 0; #10;
$display("%b %b | %b %b %b %b %b %b %b",
A,B,AND_OUT,OR_OUT,NOT_OUT,NAND_OUT,NOR_OUT,XOR_OUT,XNOR_OUT);

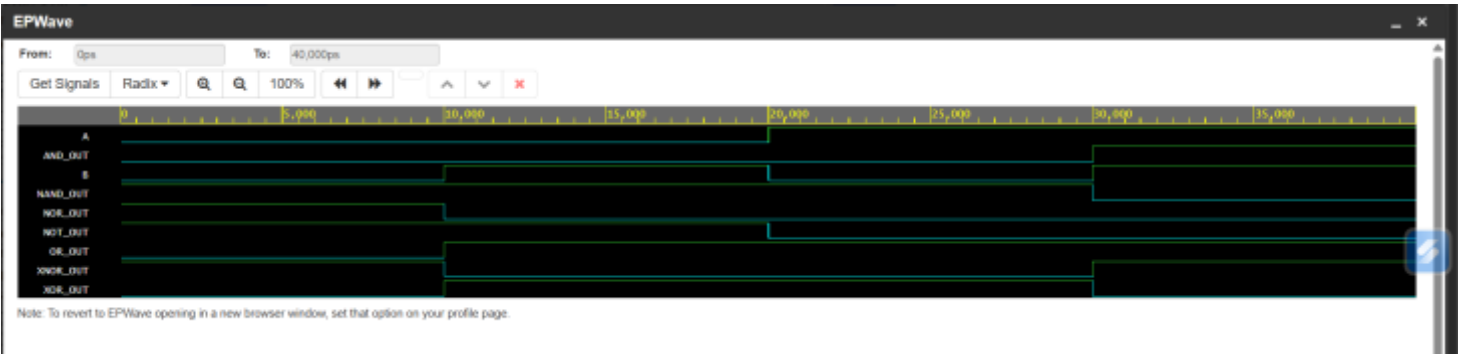
A = 1; B = 1; #10;
$display("%b %b | %b %b %b %b %b %b %b",
A,B,AND_OUT,OR_OUT,NOT_OUT,NAND_OUT,NOR_OUT,XOR_OUT,XNOR_OUT);

$finish;
end

endmodule

```

SIMULATION RESULTS:



CONCLUSION:

In this experiment, the truth tables of various logic gates were verified by applying all possible input combinations and observing the outputs. The practical results matched the theoretical truth tables of AND, OR, NOT, NAND, NOR, XOR, and XNOR gates. This confirms the correct working of the logic gates. The experiment helped in understanding the logical operations and behavior of digital circuits. Hence, truth table verification is essential for designing and analyzing combinational logic systems.